Explainable machine learning for automated robotic surgical skill assessment using rich kinematic features.

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Abstract

Robotic-assisted surgery (RAS) extends minimally invasive surgery by restoring dexterity, tremor filtration, and ergonomic console control compared with open procedures and conventional laparoscopy. Objective, scalable skill assessment from console kinematics is therefore clinically relevant for competency-based training. Existing JIGSAWS studies often rely on random trial splits, isolated handcrafted features, or black-box models without subject-independent validation or transparent explanations. This study presents an explainable machine learning framework for automated skill classification from kinematic data alone. Using the public JHU-ISI Gesture and Skill Assessment Working Set (JIGSAWS), comprising 103 trials from 40 subjects across three dry-lab tasks, 633 trial-level kinematic features were extracted and six supervised classifiers were compared under leave-one-subject-out cross-validation (LOSO-CV). Random Forest (RF) with feature standardization was selected as the primary model; interpretability was assessed with SHapley Additive exPlanations (SHAP). Key gaps addressed include: subject-independent evaluation on all trials, systematic classifier comparison on identical features, collinearity-aware feature analysis, global and local SHAP attribution, and ablations (mean/standard-deviation features, segment-level modeling, PCA reduction, deep sequences, cross-task transfer). Under pooled LOSO-CV, RF with advanced kinematic features achieved 97.1% accuracy and Cohen's [Formula: see text] (macro-F1 [Formula: see text]; balanced accuracy [Formula: see text]), outperforming PCA-reduced features (92.2%) and segment-level modeling (88.3%). SHAP analysis identified jerk and velocity variability in master-tool channels as dominant discriminators. Cross-task evaluation showed strong in-task performance (97-100%) but reduced off-task transfer (61-95%). These benchmark findings may inform explainable kinematics-based training analytics; external multi-center validation remains required before clinical deployment.

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